

MINNEAPOLIS ANOKA COUNTY AP, MN, USA

WMO: 726577

Lat: 45.150N

Lon: 93.217W

Elev: 278

StdP: 98.03

Time Zone: -6.00 (NAC)

Period: 97-19

WBAN: 94974

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.7	-20.3	-27.3	0.3	-22.2	-24.9	0.4	-20.5	10.6	-6.9	9.3	-9.3	2.8	290	0.561

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.1	32.3	23.7	30.9	23.1	28.8	22.0	25.6	29.9	24.3	28.5	23.3	27.3	5.4	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.0	19.5	28.2	22.8	18.1	26.8	22.2	17.4	26.2	79.6	30.4	74.7	28.6	70.6	26.9	29.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 10.3	8.8	7.9		-26.1	34.7	3.3	1.4	-28.5	35.7	-30.4	36.5	-32.2	37.3	-34.6	38.3	
(5)			DB	-26.4	27.3	3.1	1.4	-28.6	28.4	-30.4	29.2	-32.1	30.0	-34.4	31.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.8	-8.9	-4.8	-0.6	8.1	13.7	19.2	22.3	20.8	17.1	9.6	1.9	-5.2
	DBStd	11.87	6.71	6.42	6.36	5.50	4.57	4.07	3.50	3.23	4.89	5.35	5.62	6.60
	HDD10.0	2246	585	414	332	98	17	0	0	0	4	74	251	472
	HDD18.3	4180	843	648	586	311	157	37	7	13	79	276	493	729
	CDD10.0	1456	0	0	4	41	131	277	381	335	218	62	7	0
	CDD18.3	347	0	0	0	3	12	64	129	89	43	5	0	0
	CDH23.3	3087	0	0	0	42	143	569	1203	732	365	33	0	0
	CDH26.7	947	0	0	0	14	42	164	415	202	104	6	0	0
(14) Wind	WSAvg	3.5	3.5	3.6	3.7	4.0	3.8	3.3	2.9	2.8	3.4	3.6	3.6	3.5
(15) Precipitation	PrecAvg	837	22	20	42	75	103	121	117	108	84	71	47	27
	PrecMax	1116	51	43	93	201	228	222	237	192	198	157	137	55
	PrecMin	622	2	4	12	20	4	47	23	10	10	18	1	5
	PrecStd	123	13	10	20	38	52	45	51	48	51	42	36	13
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.9	11.2	19.0	28.7	31.2	32.8	34.0	33.8	32.1	27.3	20.8	12.7
		MCWB	4.5	7.3	12.4	19.4	20.5	23.5	24.1	25.3	22.5	20.0	12.9	9.5
	2%	DB	3.8	7.8	16.0	22.7	27.4	30.9	32.4	31.0	29.1	23.0	17.1	7.7
		MCWB	1.7	4.9	11.8	13.6	17.7	22.7	24.2	23.9	21.4	16.6	12.8	5.2
	5%	DB	2.1	6.2	12.4	20.2	23.9	28.0	31.1	28.7	27.5	21.0	13.0	5.9
		MCWB	1.1	4.0	8.3	12.2	15.4	20.8	23.7	22.5	20.1	15.8	9.2	3.4
	10%	DB	0.0	3.9	8.8	17.6	22.3	27.1	28.9	27.4	25.2	17.8	10.3	2.8
		MCWB	-0.6	2.3	6.1	10.5	15.1	19.9	22.8	21.0	18.7	13.5	7.0	1.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.0	7.5	14.6	19.6	22.0	25.6	27.5	27.5	24.2	21.3	16.3	9.8
		MCDB	7.9	9.3	16.8	26.9	28.4	30.8	31.9	31.0	29.4	24.0	17.0	13.7
	2%	WB	2.3	5.6	11.7	15.8	19.2	23.9	25.9	25.0	22.8	18.4	13.1	5.9
		MCDB	3.9	7.7	15.3	20.7	24.7	28.4	30.0	28.7	26.9	21.7	16.2	7.8
	5%	WB	0.8	4.2	9.1	12.9	17.5	22.6	24.8	23.5	21.6	16.7	9.9	3.2
		MCDB	1.7	6.1	12.2	18.5	22.1	26.8	28.9	27.5	26.1	19.1	13.4	5.3
	10%	WB	-0.9	2.6	5.6	11.3	15.8	21.3	23.7	22.2	20.1	14.5	7.4	1.5
		MCDB	-0.2	4.2	8.9	16.5	20.3	25.3	27.8	26.0	24.2	17.8	9.9	3.0
(35) Mean Daily Temperature Range	5% DB	MDBR	8.1	8.5	9.3	11.1	10.8	10.4	10.1	10.5	11.0	10.1	8.2	7.7
		MCDBR	8.4	8.8	13.4	15.2	14.2	11.7	11.5	11.8	12.5	13.0	12.8	10.3
		MCWBR	7.0	6.5	9.1	8.3	7.1	5.7	5.3	6.0	6.3	7.8	8.3	7.7
	5% WB	MCDBR	8.2	8.2	12.6	13.2	12.5	10.2	10.1	10.3	11.2	10.5	11.6	9.4
		MCWBR	7.1	6.3	9.1	8.3	7.7	6.1	5.6	6.5	6.4	7.4	8.6	7.4
(40) Clear-Sky Solar Irradiance	taub		0.268	0.280	0.319	0.379	0.409	0.399	0.395	0.404	0.375	0.321	0.291	0.268
	taud		2.410	2.361	2.376	2.286	2.250	2.328	2.359	2.337	2.385	2.529	2.514	2.433
	Ebn at Noon		857	910	914	881	861	870	868	846	843	855	826	818
	Edh at Noon		71	91	104	125	133	124	119	117	103	77	65	63
(44) All-Sky Solar Radiation	RadAvg		1.72	2.81	3.83	4.61	5.33	5.92	6.27	5.23	4.07	2.57	1.63	1.20
	RadStd		0.20	0.28	0.34	0.45	0.43	0.46	0.22	0.38	0.28	0.34	0.15	0.14

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (61 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page